

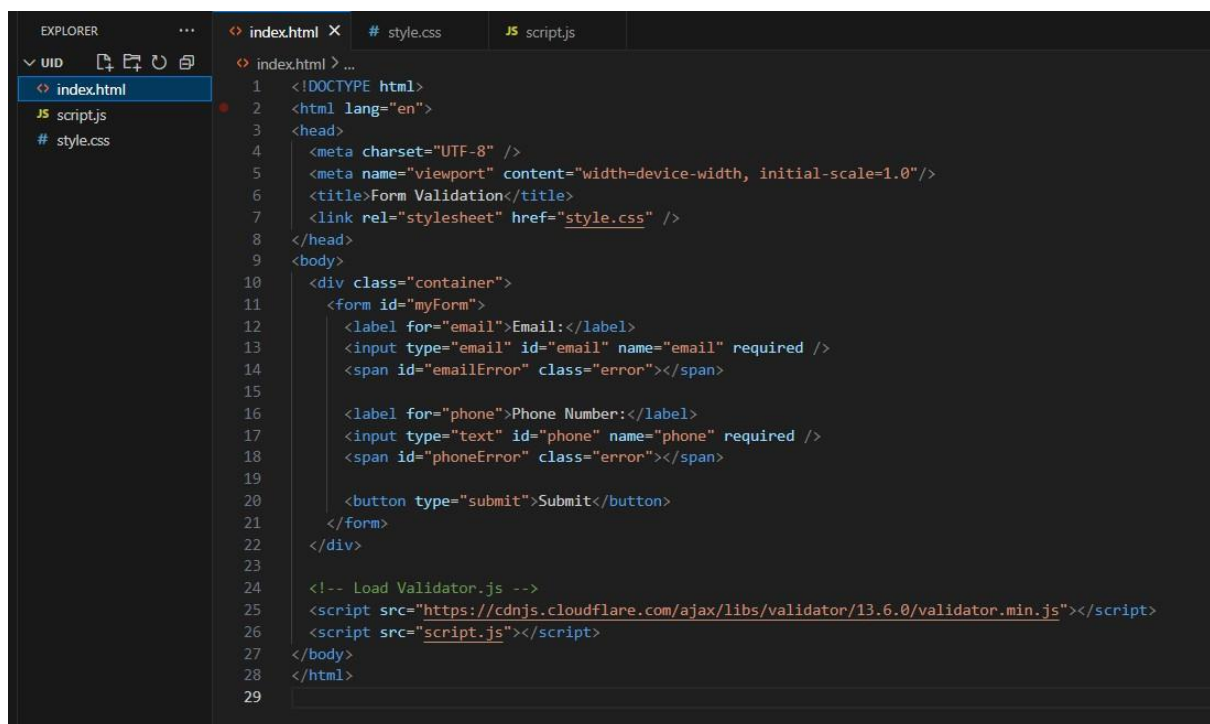
WEEK 9: DESIGN INPUT FORMS THAT VALIDATE DATA (E.G., EMAIL, PHONE NUMBER) AND DISPLAY ERROR MESSAGES USING HTML/CSS, JAVASCRIPT (WITH VALIDATOR.JS)

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AIM: The aim is to design input forms that validate data, such as email and phone number, and display error messages using HTML/CSS and JavaScript with Validator.js.

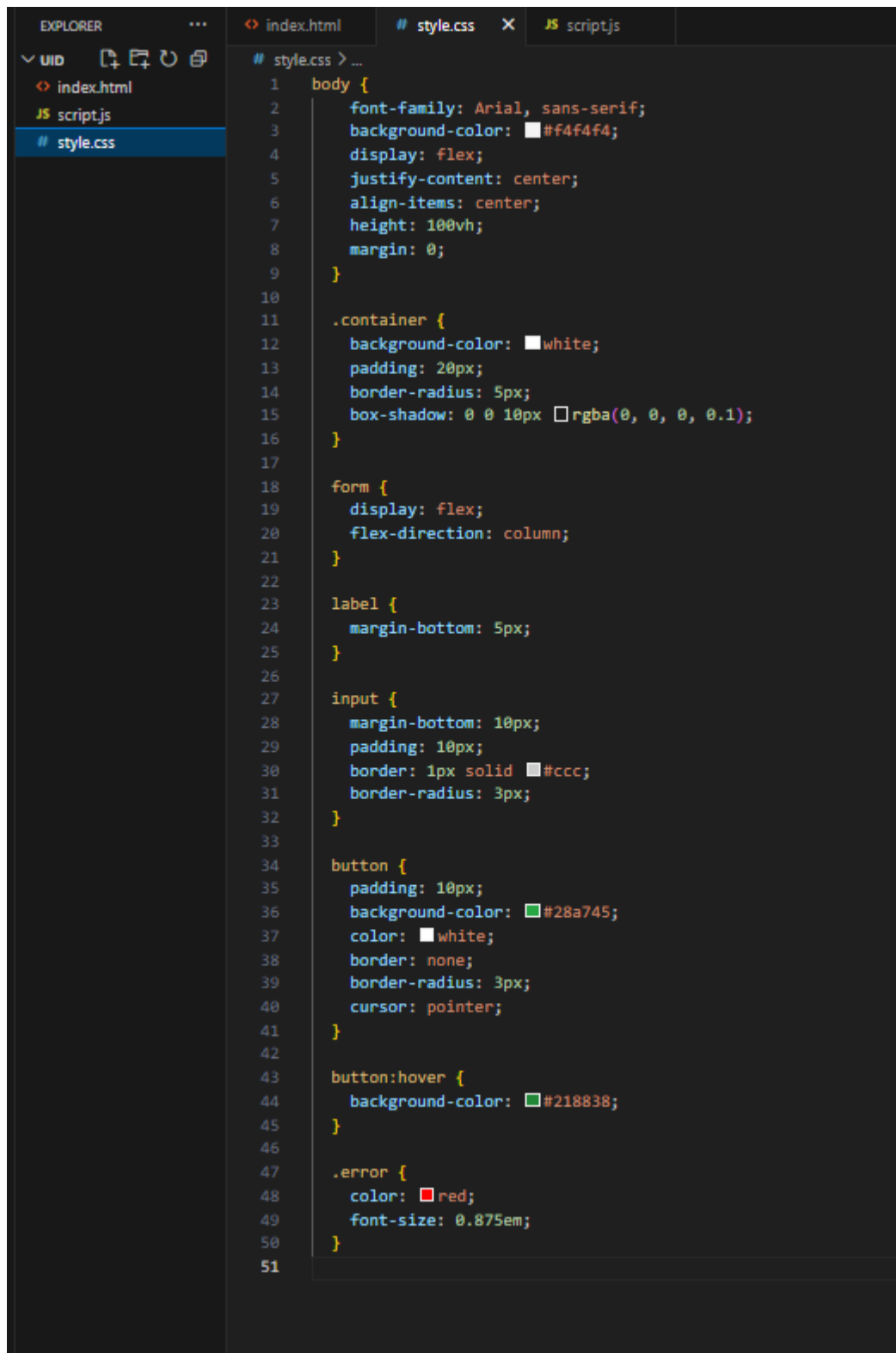
PROCEDURE:

Step 1: Setting Up the HTML Form Start by creating an HTML form with input fields for the email and phone number.

A screenshot of a code editor with a dark theme. The Explorer sidebar on the left shows a project structure with 'index.html', 'script.js', and 'style.css'. The main editor area displays the content of 'index.html'. The code includes a DOCTYPE declaration, HTML and head tags with meta and title elements, a link to 'style.css', and a body containing a form with email and phone number inputs, error spans, and a submit button. At the bottom, there is a comment to load 'Validator.js' and two script tags: one for the CDN version and one for the local 'script.js' file.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
6   <title>Form Validation</title>
7   <link rel="stylesheet" href="style.css" />
8 </head>
9 <body>
10  <div class="container">
11    <form id="myForm">
12      <label for="email">Email:</label>
13      <input type="email" id="email" name="email" required />
14      <span id="emailError" class="error"></span>
15
16      <label for="phone">Phone Number:</label>
17      <input type="text" id="phone" name="phone" required />
18      <span id="phoneError" class="error"></span>
19
20      <button type="submit">Submit</button>
21    </form>
22  </div>
23
24  <!-- Load Validator.js -->
25  <script src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.6.0/validator.min.js"></script>
26  <script src="script.js"></script>
27 </body>
28 </html>
29
```

Step 2: Styling the Form with CSS Next, add some basic styling to make the form look nice.

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project named 'UID' with files 'index.html', 'script.js', and 'style.css'. The 'style.css' file is selected and open in the main editor. The code in the editor is CSS for a form, including styles for the body, container, form, label, input, button, and error messages. The code is as follows:

```
1 body {
2   font-family: Arial, sans-serif;
3   background-color: #f4f4f4;
4   display: flex;
5   justify-content: center;
6   align-items: center;
7   height: 100vh;
8   margin: 0;
9 }
10
11 .container {
12   background-color: white;
13   padding: 20px;
14   border-radius: 5px;
15   box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
16 }
17
18 form {
19   display: flex;
20   flex-direction: column;
21 }
22
23 label {
24   margin-bottom: 5px;
25 }
26
27 input {
28   margin-bottom: 10px;
29   padding: 10px;
30   border: 1px solid #ccc;
31   border-radius: 3px;
32 }
33
34 button {
35   padding: 10px;
36   background-color: #28a745;
37   color: white;
38   border: none;
39   border-radius: 3px;
40   cursor: pointer;
41 }
42
43 button:hover {
44   background-color: #218838;
45 }
46
47 .error {
48   color: red;
49   font-size: 0.875em;
50 }
51
```

Step 3: Adding JavaScript for Validation Finally, add JavaScript to validate the input fields using Validator.js and display error messages.

```
1 document.getElementById('myForm').addEventListener('submit', function (e) {
2   e.preventDefault();
3
4   let email = document.getElementById('email').value.trim();
5   let phone = document.getElementById('phone').value.trim();
6   let emailError = document.getElementById('emailError');
7   let phoneError = document.getElementById('phoneError');
8
9   // Clear previous error messages
10  emailError.textContent = '';
11  phoneError.textContent = '';
12
13  let isValid = true;
14
15  // Validate email
16  if (!validator.isEmail(email)) {
17    emailError.textContent = 'Please enter a valid email address.';
18    isValid = false;
19  }
20
21  // Validate phone number (international format allowed)
22  if (!validator.isMobilePhone(phone, 'any')) {
23    phoneError.textContent = 'Please enter a valid phone number.';
24    isValid = false;
25  }
26
27  // If valid, log the values (or proceed to submit)
28  if (isValid) {
29    console.log('Email:', email);
30    console.log('Phone:', phone);
31    alert('Form submitted successfully!');
32    // Optionally reset the form
33    // document.getElementById('myForm').reset();
34  }
35  });
36
```

OUTPUT1:

Email:

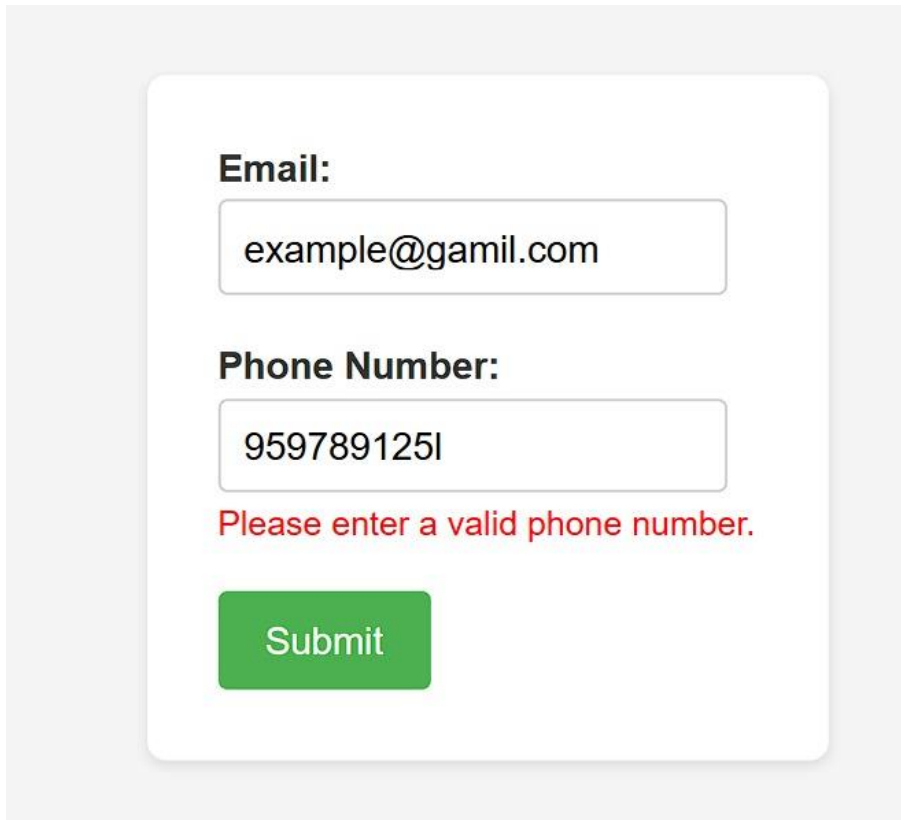
Phone Number:

Submit

Output 1 has these features in it:

- A label and input field for **Email**
- A label and input field for **Phone Number**
- A **Submit** button styled with a green background

OUTPUT 2:



Email:

Phone Number:

Please enter a valid phone number.

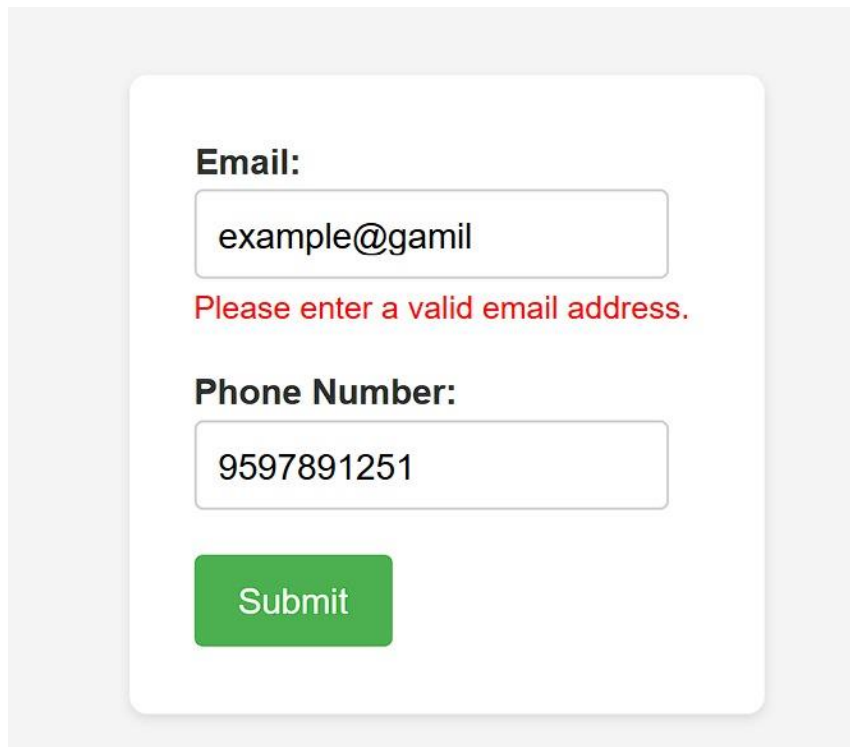
Submit

We got a error message in the above output:

"Please enter a valid phone number." (in red colour)

In the above output input 1 email is correct but input 2 the phone number is incorrect, because it must contain only digit but it also contain a alphabet therefore incorrect.

OUTPUT 3:



Email:

Please enter a valid email address.

Phone Number:

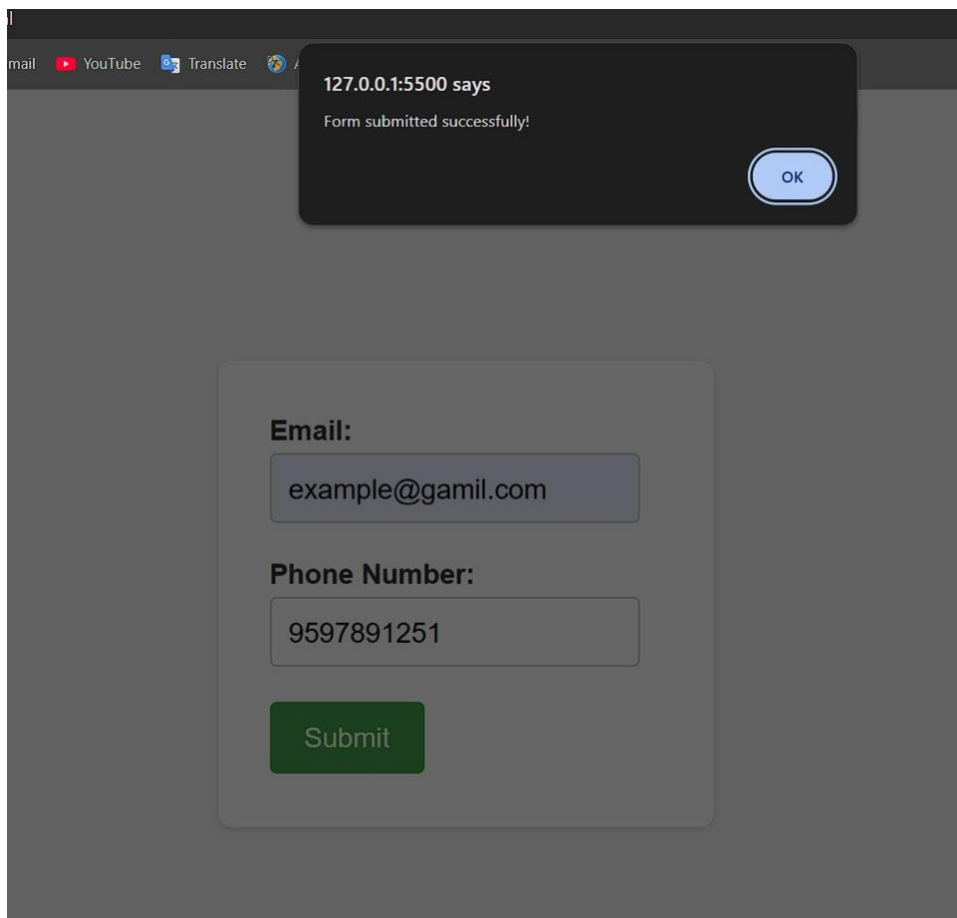
Submit

We got a error message in the above output:

"Please enter a valid email address." (in red colour)

In the above output input 2 phone number is correct but input 1 email is number is incorrect, because it must contain a valid mail id it should end with .com

OUTPUT 4:



Here the user has entered the correct details in both email id and phone number and we can see from the below image that the data is successfully processed and it has been successfully submitted as well.

